

Rakshita Vijay

The best systems are the ones people never have to think about.

[Chennai, India](#) | [GitHub](#) | [LinkedIn](#)

About

I'm a third-year Computer Science Engineering student at BITS Pilani, Hyderabad, with a habit of picking projects apart down to the data structure underneath - whether that's a dependency graph deciding the order database objects get migrated in, or a function-call graph deciding whether an Android app is malicious.

Most recently I was a Software Engineering Intern in Microsoft's Azure Data Migration team, where I extended an AI-assisted schema-conversion pipeline to handle SQL Server as a source: a topological chunk scheduler, a hybrid regex-plus-LLM translation router, and a fault-tolerant compile-and-retry loop that got ~97% of chunks converting without manual intervention. Before that, at CultureVo, I built a multi-agent conversational AI toolkit spanning six personas and four languages.

Outside of internships, I prompt-engineer AI-driven tools end to end - a finance tracker with a Gemini-powered, 2-agent insight pipeline, a project-tagging workspace on Next.js and Supabase, an async research-and-ideation pipeline that cut a team's turnaround from five days to one sitting.

When I'm not at a keyboard, I'm usually working a crochet or knitting pattern - a different kind of graph traversal - or buried in a novel, mystery and/or fantasy being my usual picks.

Education

BITS Pilani, Hyderabad Campus

2023 - 2027

B.E. Computer Science Engineering · CGPA 8.64

St. John's Public School

Class XII · 2023

CBSE · 94.2%

Chettinad Vidyashram

Class X · 2021

CBSE · 95.8%

Technical Skills

Python, C++, C, Java, SQL, PyTorch, Generative AI, Next.js, React, Supabase, Vercel, Gemini API, Prompt Engineering, Git

Experience

Microsoft - Azure Data Migration Team

May 2026 - Jul 2026

Software Engineering Intern · SQL Server → PostgreSQL Schema Conversion Engine

- Extended an AI-assisted Oracle→PostgreSQL schema-conversion pipeline to also support SQL Server as a source, with a near real-time ETL pipeline.
- Implemented dialect-aware gating across all 4 stages (Extract→Chunk→Convert→Verify), preserving Oracle behavior and aligning with the PG team.
- Built a dependency-aware chunk scheduler on Kahn's Algorithm that topologically orders database objects for conversion - discovering in-scope objects, emitting raw DDL plus a dependency edge graph, and correctly handling cycles and self-referencing edges while upholding ACID transactionality.
- Designed a hybrid conversion router that sent mechanical DDL through a deterministic regex fast-path and reserved LLM-based translation for procedural/semantic logic, using few-shot examples and a shared canonical identifier map to keep naming consistent across both paths - optimizing token usage.
- Built a fault-tolerant compile-and-retry loop that test-compiles each chunk's DDL, defers cross-chunk dependencies, and auto-retries failures from compile-error feedback - ~97% of chunks converged without manual intervention.
- Stress-tested end-to-end against real-world corpora (Northwind, AdventureWorks, WideWorldImporters) plus a custom 7-schema database: ~98% of in-scope objects converted across 45+ chunks (600+ objects with dependencies), cutting runtime from ~4 hours to 35-45 minutes (~83% reduction), correctly identifying and excluding known out-of-scope objects via the fix-loop.

CultureVo

May 2025 - Jul 2025

Summer Intern · Conversational AI Tools

- Built PersonaBot, a configurable multi-agent orchestration toolkit with a state-managed persona system spanning 6 profiles and 10 customizable traits.
- Implemented persistent, multi-session conversation state across four languages (English, Sinhala, Tamil, Hindi), enabling bulk generation of 100+ Q&A pairs.

Projects

Android Malware Detection with GNNs (GraphSAGE, Androguard, PyTorch)

Jan 2026 - May 2026

Cybersecurity · Graph ML

- Built an end-to-end malware classification pipeline extracting Function Call Graphs (FCGs) via Androguard and training GraphSAGE models to distinguish benign from malicious Android apps.
- Designed an API-aware, structure-preserving graph pruning technique that removes low-signal non-API methods via weighted edge rewiring.
- Integrated pretrained Skip-Gram node embeddings for graph preprocessing/serialization on HPC, with checkpoint-based training for resumable runs.
- Achieved a 94.39% F1-score and 92.41% accuracy on an imbalanced dataset of 150K benign and 50K malicious samples.

Finance Tracker & Analyser (Next.js, Gemini 3.5 Flash, Vercel)

Jun 2025 - Jul 2025

AI-Driven Financial Analytics

- Designed a multi-format analysis and report-generation service using Gemini 3.5 Flash, backed by a 2-agent pipeline for automated insight generation.
- Implemented real-time budget tracking with configurable spending categories, plus an AI-driven module for recovery-strategy suggestions.
- Built a rules-based anomaly-detection service that processes uploaded statement data and flags inconsistencies for user review.

Automated Research & Ideation Pipeline (Python, asyncio, Multi-Agent)

Jun 2025

Multi-Agent Systems · Async Automation

- Designed a 5-agent orchestration workflow to automate topic selection, research, and outline generation, coordinating task handoffs and shared agent state.
- Implemented concurrent agents over Python asyncio to generate 7 niche titles per theme, each with research summaries, sources, and article outline guides.
- Cut research-and-ideation turnaround by ~80% (5 days → 1 session) with compiled, referenced sources; adopted as the team's standard ideation tool.

Tag Atlas (Next.js, Supabase, Gemini)

Personal project

Project write-up storage & tagging tool

- Sidebar-plus-detail-panel workspace for storing and tagging project write-ups, with tag-based similarity ranking and an archive flow.
- AI-assisted tag suggestions via Gemini, password auth, and bulk project-insertion tooling.

Leadership

ACM - BPHC Chapter

Nov 2024 - Jul 2026

Content & Research Team Co-Lead, Junior Office & Chair, Senior Office

- Authored 5+ technical articles for the ACM BPHC Blog and The BITSian Gazette, BITS Hyderabad's official magazine (Aug–Nov '24 edition).
- Conducted and coordinated logistics as Point of Contact for 4 tech events, including a coding workshop for 70+ indigent students.

ELAS, BPHC

Dec 2024 - Jan 2025

ARG Team Lead, Verba Maximus '25 · English Language Activities' Society (ELAS)

- Led a 13-member team to design and execute a 6-hour event that drew 125+ participants - the highest-footfall headliner event at VM '25.

Talks & Workshops

Introduction to Machine Learning

Aug 2025

Organized by Association of Computing Machinery - BPHC Chapter

- Co-hosted a 250+ attendee workshop covering supervised/unsupervised learning, regression, neural networks, and activation functions with live coding.